Cheatsheet AZ-305 — 06 · Messaging and events

1. The four services

	Queue Storage	Service Bus	Event Hubs	Event Grid
Type	Simple queue	Enterprise broker (queues + topics)	Event streaming	Push notifications
What it carries	Small messages (64 KB)	Messages (256 KB / 100 MB Premium)	Massive data/telemetry streams	Discrete events ("something happened")
Model	Pull	Pull (queues) / pub-sub (topics)	Pull, partitioned, consumer groups	**Push** to subscribers
Volume	Very high (up to 500 TB)	High	**Millions/sec**	Individual events
Guaranteed order (FIFO)	✗ (best effort)	✓ with **sessions**	Per partition	✗
Transactions / duplicates / dead-letter	✗	✓ (dedup, DLQ, transactions)	✗	24 h retries, optional dead-letter
Retention	Up to 7 days	Until configurable expiry	1–7 days (more in Premium/Dedicated), **replayable**	24 h retries
Protocols	REST	AMQP, REST	AMQP, Kafka, REST	HTTPS, webhooks, CloudEvents
Typical producers	Apps	Apps	Apps, devices, **diagnostic settings**	Azure resources (Blob, ARM, Key Vault, IoT Hub), custom topics
Typical consumers	Apps, Functions	Apps, Functions, Logic Apps	Stream Analytics, Functions, Databricks, Synapse	Functions, Logic Apps, webhooks, Event Hubs
Typical case	Cheap queue between microservices	Orders, FIFO, pub/sub, integration	IoT telemetry, logs, clickstream	"When a blob is created, run X"

2. Service Bus: exam features

Feature	What it does	Signal
Sessions	Groups messages by SessionId; **guaranteed FIFO** and one consumer per session	"FIFO", "process in order"
Partitioning	Spreads the queue across brokers → more throughput; **breaks global order**	"throughput", never "FIFO"
Topics + subscriptions	Pub/sub with filters per subscription	"multiple subscribers", "filter"
Auto-forwarding	Chains topic/subscription → destination queue	"forward to individual queues"
Dead-letter queue	Messages that cannot be processed	"poison messages"
Duplicate detection	Discards duplicates by MessageId	"exactly once"
Scheduled delivery / deferral	Deliver later / postpone	"delay"
Premium	Dedicated resources, VNet/Private Endpoint, large messages	"isolation", "large messages"

3. Event Grid vs Event Hubs (the usual confusion)

Question	Event Grid	Event Hubs
Who produces?	Azure resources that support it natively, or your app (custom topic)	Apps, devices, **diagnostic settings of any service (incl. Entra ID)**
What do I want to do?	**React** to each event	**Ingest, retain and analyze** a stream
Size / frequency	Small, sporadic	Large, continuous
Mental phrase	"Tell me when something happens"	"Give me everything that happens, raw"

- **Entra ID does not publish to Event Grid.** Its logs only come out via **diagnostic settings** → Log Analytics / Storage / **Event Hubs**.
- Pipeline "Entra ID → ? → ? → Cosmos DB": **Event Hubs → Functions → Cosmos DB**.
- **Event Grid runs C# code with access to a private IP + minimum cost → Functions Premium** (Consumption has no VNet integration; Logic Apps ISE is very expensive).

4. Diagnostic settings: the three destinations

Destination	What for	Signal
Log Analytics	Query (KQL) and **alert**	"analyze", "alerts"
Storage account	Cheap long-term archiving	"retain for 7 years", "archive"
Event Hubs	**Streaming** to SIEM / external system / pipeline	"third-party SIEM", "stream", "Cosmos DB"

5. Other integration services

Service	What it is	Signal
Logic Apps	Low-code orchestration with connectors; Consumption vs Standard (VNet) vs ISE (isolated, expensive)	"no code", "connectors", "workflow"
Functions	Code in response to events; Consumption / Premium (VNet, no cold start) / Dedicated	"custom code", "C#"
Durable Functions	Stateful workflows; **Orchestrator** defines the flow and the **durable timers**; Activity does the work; Client starts it	"durable timer" → Orchestrator
On-premises Data Gateway	Logic Apps / Power BI / Power Automate access on-prem SQL without opening ports	"no inbound ports"
Self-hosted Integration Runtime (formerly Data Management Gateway)	Data Factory / Synapse access on-prem data	"ADF + on-premises"
Notification Hubs	Push to mobile (iOS/Android)	"mobile push"
SignalR Service	Real time to browsers	"real-time web"
Azure Cache for Redis	In-memory cache, distributed **session state** (serializable), close to the app	"session state", "serializable", "near the app"
CDN	Static content close to the **user**	"near the user"

6. Decision tree

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What do I need to move?
├─ A notice that something happened in Azure, to react ....... Event Grid
├─ A massive data stream to analyze / retain ................. Event Hubs
├─ Entra ID logs / diagnostic settings to the outside ........ Event Hubs
├─ Messages with FIFO / transactions / pub-sub / DLQ ......... Service Bus (sessions for FIFO)
├─ Simple, cheap, REST queue, no guarantees .................. Queue Storage
├─ Push to mobile ............................................ Notification Hubs
└─ Scalable web session state ............................... Azure Cache for Redis
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7. Signals → answer

Wording	Answer
"FIFO"	Service Bus + sessions
"forward to individual queues from a topic"	Auto-forwarding
"simple REST queue between microservices"	Queue Storage
"user creation and role assignment events → Cosmos DB", service 1	Event Hubs (service 2: Functions)
"blob created → process it"	Event Grid
"IoT telemetry, millions of events"	Event Hubs
"code in response to Event Grid + private IP + min cost"	Functions Premium
"durable timer"	Orchestrator function
"Logic App → SQL on-prem, no inbound ports"	On-premises Data Gateway
"session state, serializable, scalable"	Azure Cache for Redis (clustered)

8. Mistakes to avoid

1. Event Grid for Entra ID logs → it is not a diagnostic settings destination.
2. Partitioning for FIFO → breaks the order; sessions.
3. Queue Storage when order or transactions are required.
4. Functions Consumption to reach a private IP.
5. Output Cache provider confused with Session State provider (Redis).